

CROSS-E-strong-neg-02 v1: W- control bottleneck for E-strong

ZFL Lab

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Classification: $C_E^{weak} = \text{PROVEN } C_{L^3} = \text{PROVEN } C_E^{strong} = \text{OPEN MISSION}$: Audit whether W_- is temporally controlled enough to imply C_E^{strong} . SUCCESS: integrability / domination of $W_- =_i C_E^{strong}$ PROVEN FAILURE: no such control $=_i$ OPEN / BLOCKED FORBIDDEN: $|W|$ bound $=_i W_-$ integrable; W_+ divergent $=_i$ cumulative divergent; ESS $=_i E^{strong}$ without bridge.

Setup:

$W(t) := \int_{R^3} \omega^T S \omega dx$ signed. Decompose $W(t) = W_+(t) - W_-(t)$, $W_{\pm}(t) \geq 0$ pointwise in time via positive/negative part of density, or $W_+ = \max(W, 0)$, $W_- = \max(-W, 0)$.

From C_E^{weak} PROVEN: $T_* < \infty \implies \limsup_{t \rightarrow T_*} \int_0^t W(\tau) d\tau = +\infty$.

Since $\int W = \int W_+ - \int W_-$, $\limsup \int W = +\infty$ implies

$$\int_0^{T_*} W_+(t) dt = +\infty.$$

Indeed, if $\int W_+$ were finite, $\limsup \int W \leq \int W_+ < \infty$, contradiction. So W_+ non-integrable is consequence of C_E^{weak} .

Goal for C_E^{strong} : $\lim_{t \rightarrow T_*} \int_0^t W = +\infty$.

Write $\int_0^t W = \int_0^t W_+ - \int_0^t W_-$.

Obstacle is $\int_0^t W_-$.

Target controls:

(A) Strong:

$$\boxed{\int_0^{T_*} W_-(t) dt < \infty}$$

Then $\int_0^t W \rightarrow +\infty$ because $\int W_+ \rightarrow +\infty$ and $\int W_-$ bounded.

(B) Weak domination:

$$\boxed{\int_0^t W_-(\tau) d\tau = o\left(\int_0^t W_+(\tau) d\tau\right), t \rightarrow T_*}$$

Then $\int W = (1 - o(1)) \int W_+ \rightarrow +\infty$.

Hence if (A) or (B) proven, C_E^{strong} PROVEN.

Why current bound insufficient:

We have $|W| \leq C_S \|u\|_3 \|\nabla \omega\|_2^2$. Thus $0 \leq W_- \leq |W| \leq C_S \|u\|_3 \|\nabla \omega\|_2^2$. Near T_* , $\|u\|_3$ may blow up in limsup (ESS) and $\|\nabla \omega\|_2^2$ is expected to be non-integrable if enstrophy blows up. Product

bound gives upper bound diverging, not integrable control. So: $|W|$ bound does NOT imply $W_- \in L_t^1$.

Similarly, ESS $T_* < \infty \implies \limsup \|u\|_3 = \infty$ does not control $\int W_-$.

Thus W_- temporal control is independent hypothesis.

Result of audit:

From hypotheses $\{C_E^{weak}, C_{L^3}, |W| \leq C_S \|u\|_3 \|\nabla \omega\|_2^2\}$:

$$\int_0^{T_*} W_- < \infty \quad \text{NOT PROVEN}$$

$$\int W_- = o\left(\int W_+\right) \quad \text{NOT PROVEN}$$

Therefore:

$$C_E^{strong} \text{ remains OPEN / BLOCKED by } W_- \text{ control}$$

No contradiction, no implication fabricated.

State remains $\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$.

Next step would require new estimate: e.g., bound on negative part of strain $\|S_-\|$ or geometric depletion that makes $W_- \ll W_+$ near blow-up.